

EX NO: 2	Shell Programming Using Control Statements
Date: 17-06-26	

AIM:

To evaluate the use of shell programming control structures.

DESCRIPTION:

(a) if . . . fi structure :

The if . . fi structure allows shell to make decisions and execute statements conditionally.

SYNTAX: if [expression] then statements

fi

(b) if . . . else statement

If else statements can be used to select an option from a given set of options.

SYNTAX: if [expression] then

statement(s) else statement

fi

(c) while loop:

The while loop enables us to execute a set of commands repeatedly until some condition occurs.

SYNTAX:

while [condition] do statement(s)

done

(d) for loop

The for loop operates on list of items it repeats a set of commands for every item in a list.

SYNTAX:

for((i=0; i<n; i++))

do statement(s)

done

(a). To check whether even or odd

Input: CODE:

```
echo "Mitun 25IT371";
```

```
echo "Enter a no:";
```

```
read n;
```

```
a=$((n%2));
```

```
if [ $a -eq 0 ];
```

```
then
```

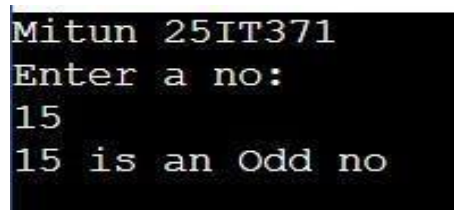
```
echo "$n is an Even no";
```

```
else
```

```
echo "$n is an Odd no";
```

```
fi
```

Output:



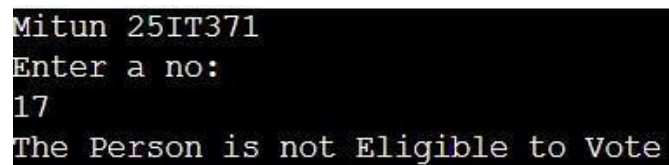
```
Mitun 25IT371
Enter a no:
15
15 is an Odd no
```

(b). To check whether a person is eligible to vote

Input: CODE:

```
echo "Mitun 25IT371";  
echo "Enter a no:";  
read n;  
if [ $n -lt 18 ];  
then  
echo "The Person is not Eligible to Vote";  
else  
echo "The Person is Eligible to Vote";  
fi
```

Output:



```
Mitun 25IT371  
Enter a no:  
17  
The Person is not Eligible to Vote
```

(c). To find the sum of n natural numbers

Input: CODE:

```
echo "Mitun 25IT371";  
echo "Enter a no:";  
read n;  
sum=0;  
i=1;  
while [ $i -le $n ];  
do  
sum=$((sum+i));  
i=$((i+1));
```

```
done;  
echo "The sum of Natural nos is: $sum";
```

Output:

```
Mitun 25IT371  
Enter a no:  
10  
The sum of Natural nos is: 55
```

(d). To find the Fibonacci Series

Input: CODE:

```
echo "Mitun 25IT371";  
echo "Enter a no:";  
read n;  
a=0;  
b=1;  
echo "The Fibonacci Series is:";  
for ((i=0; i<n; i++))  
do  
echo -n "$a "  
c=$((a+b));  
a=$b  
b=$c  
done
```

Output:

```
Mitun 25IT371
Enter a no:
8
The Fibonacci Series is:
0 1 1 2 3 5 8 13
```

(e). To find the Armstrong number**Input: CODE:**

```
echo "Mitun 25IT371";
echo "Enter a no:";
read n;
sum=0;
a=$n;
len=${#n};
while [ $a -gt 0 ];
do
d=$((a%10));
sum=$((sum + d**len));
done;
if [ $sum -eq $n ];
then
echo "$n is an Armstrong no";
else
echo "$n is not an Armstrong no";
fi
```

Output:

```
Mitun 25IT371
Enter a no:
153
153 is an Armstrong no
```

(f). To find the greatest of 3 numbers

Input: CODE:

```
echo "Mitun 25IT371";  
echo "Enter no 1 2 and 3: ";  
read a b c;  
if [ $a -gt $b ] && [ $a -gt $c ];  
then  
echo "$a is greatest among 3";  
elif [ $b -gt $c ] && [ $b -gt $a ];  
echo "$b is greatest among 3";  
else  
echo "$c is greatest among 3";  
fi
```

Output:

```
Mitun 25IT371  
Enter 1 2 and 3:  
15 17 16  
17 is greatest among 3
```

(g). To find the factorial of a given number

Input: CODE:

```
echo "Mitun 25IT371";  
echo "Enter a no:";  
read n;  
fact=1;  
for ((i=1; i<n; i++))  
do  
fact=$((fact*i));  
done;  
echo "The factorial of $n is $fact";
```

Output:

```
Mitun 25IT371  
Enter a no:  
7  
The factorial of 7 is 5040
```

RESULT:

Thus the use of control structures by shell programming were constructed and demonstrated successfully.